Lab 5: Enterprise Windows Server Administration - Part II

CNIT 24200-003

Group 13

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BUSINESS CASE

Apex Technologies Corporation required a further expansion of the virtualized infrastructure for centralized management of server updates across the domain, distributed file access, remote access, and web-based server management. In terms of the organization's structure, six departments were maintained across the domain with the Information Technology (IT) department, Legal department, Research department, Finance department, Marketing department, and Sales department. Each department had three specialized security groups, with two users per group, which resulted in 18 security groups and 36 users across the organization. The virtualized infrastructure ran on two Nutanix Community Edition (CE) Acropolis Hypervisor (AHV) clusters — G13NUTXSRV01 and G13NUTXSRV02 — with domain controllers, client workstations, and Nutanix Prism Central deployed as Virtual Machines (VMs) across both clusters.

Still, several operational flaws remained. Client workstations and domain controllers had no centralized update management, which left Apex Technologies' infrastructure vulnerable to cybersecurity threats because of unpatched updates. Also, roaming profiles and redirected folders were served from exclusively G13SRV01VM, which was not recommended for a large-scale system as a single point of failure. No web-based method of server management existed for remote server administration. Also, PowerShell remoting was not validated across domain servers.

The expansion required applications and operating systems. Microsoft Windows Server 2025 was the operating system across all domain controllers and the G13SRV07 client with Windows Server Update Services (WSUS) server. WSUS was deployed for updates to be pulled from

Windows 11 and Windows Server 2025 clients to G13SRV07. Windows Admin Center was installed on G13SRV06 for web-based server management from domain servers and external environments. Distributed File System (DFS) namespaces and DFS replication were implemented via Server Manager across both Active Directory sites.

Active Directory data included Group Policy Objects (GPOs) for WSUS client configuration, roaming profile paths, and folder redirection paths pointed to the forest-wide DFS namespace. Network data included the DFS namespace path \\group13.c24200.cit.lcl\G13DFS and static Internet Protocol (IP) assignment for G13SRV07 at 44.38.13.34. WSUS managed updates for Windows 11 and Windows Server 2025 servers. DFS managed roaming profile and redirected folder data replicated between both Active Directory sites. Windows Admin Center managed server connection configurations and credentials for all domain servers and client workstations.

The physical diagram illustrated Apex Technologies Corporation's infrastructure before the planned expansion. The system stayed connected with the Computer and Information Technology (CIT) network and ran on two Nutanix CE AHV clusters, named G13NUTXSRV01 and G13NUTXSRV02. The environment had the domain controllers (G13SRV01VM, G13SRV03VM, G13SRV06), Windows 11 client workstations (G13S1WIN11VM, G13S2WIN11, G13S3WIN11VM, and G13WIN11), G13TRUENAS TrueNAS server, Prism Central, CIT network printer, CIT Domain Name System (DNS) server, and CIT Network Time Protocol (NTP) server. The former infrastructure supported Apex Technologies Corporation's needs, but within the network existed systematic issues with unmanaged updates, remote administration, control of print jobs, and inconsistent data, which was why Apex Technologies Corporation needed a cleaner setup with less manual work and better management between sites.

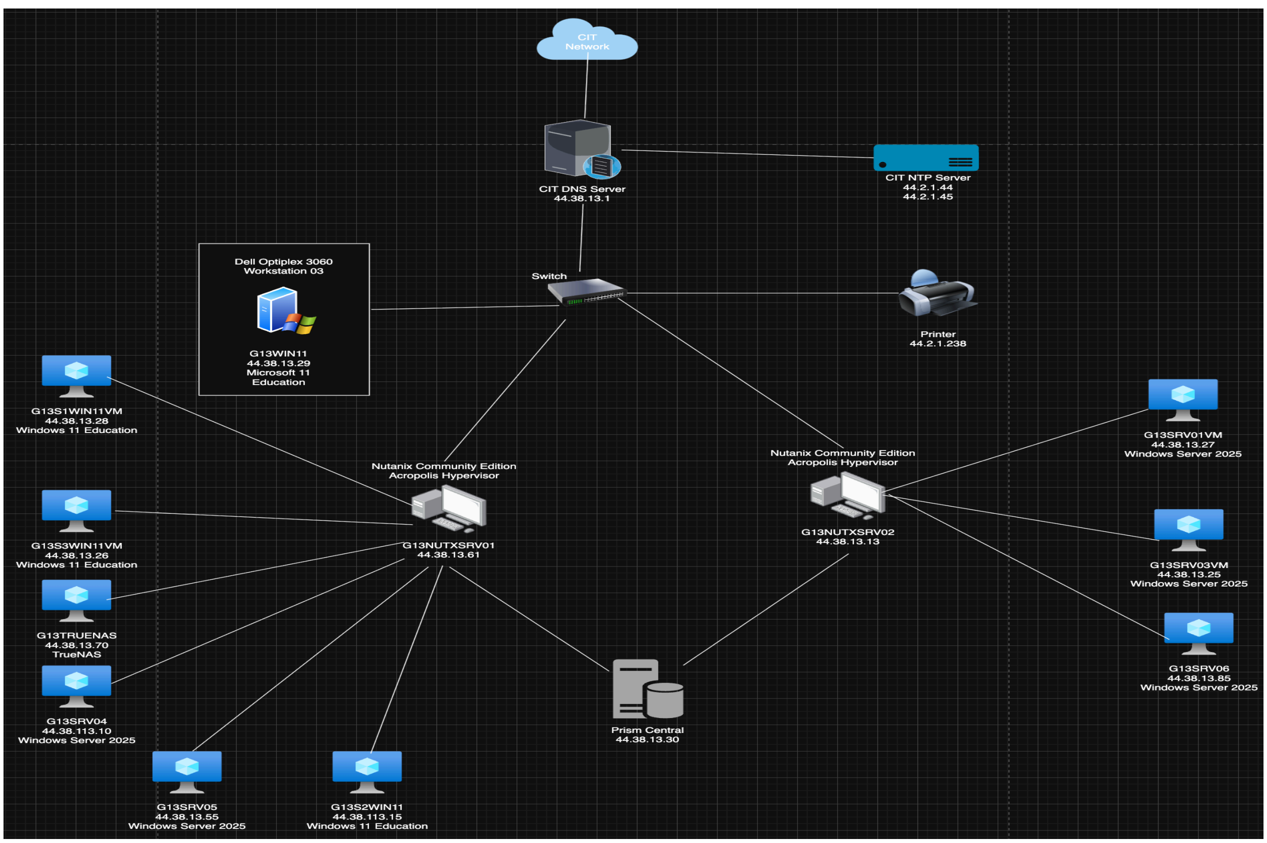

Figure 1: Physical Diagram of former Apex Technologies Corporation systems

The logical diagram represented the Active Directory structure of Apex Technologies Corporation. The root domain group13.c24200.cit.lcl contained three Organizational Units (OUs), which contained IT Department, Legal Department, and Research Department. The child domain c242-13-a.group13.c24200.cit.lcl contained three OUs (Finance Department, Sales Department, and Marketing Department). The child domain was connected to the root domain through a transitive trust, which allowed authentication and resource access across all departments. The default site on the 44.38.13.0/24 subnet contained both domains, while the LAN1913 site on the 44.38.113.0/24 subnet only contained the root group13.c24200.cit.lcl. The default IP site link connected both sites for replication traffic. The separation of domain sites ensured resources on the LAN1913 site could be accessed by only departments with high authorization levels.

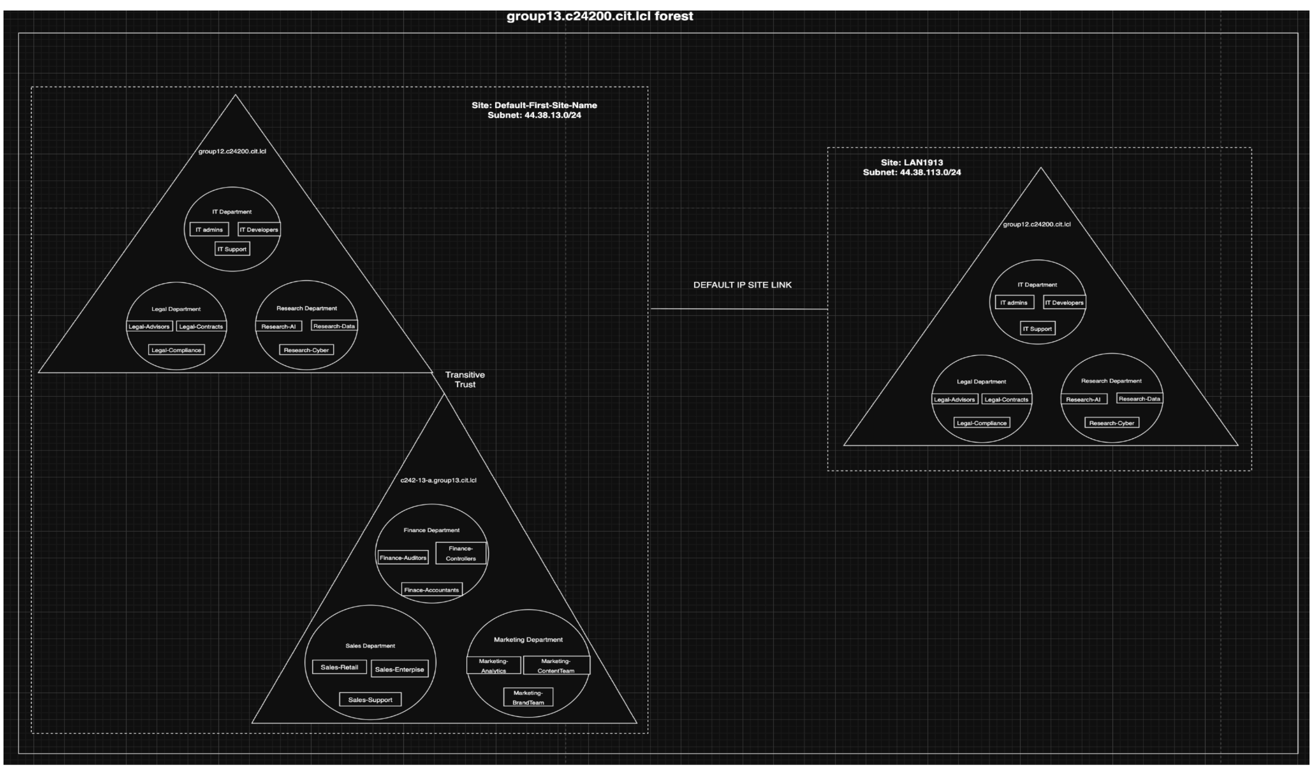

Figure 2: Logical Diagram of former Apex Technologies Corporation systems

PROCEDURES

The main objective was the continued expansion of Apex Technologies Corporation's enterprise environment through the implementation of network print services, PowerShell remoting, DFS, and WSUS. Two print queues were configured for the print server. One was for double-sided print jobs, and one was for general print jobs. One new user was created for the assignment of the print operator role and assigned a print queue with a higher priority than other users on the system. DFS was integrated across two Active Directory sites, with DFS replication enabled for a copy of the changes to the DFS structure across domain controllers. GPOs were updated for the reflection of the newly created DFS locations. WSUS was deployed on G13SRV07, and Windows Admin Center was made accessible both within the domain environment and the external environment. Clients were then configured for the acquisition of updates from the server with WSUS, rather than from Windows directly. PowerShell remoting was also enabled for remote configurations to machines across the group13.c24200.cit.lcl domain.

All **buttons** were bold, *options* were italicized, `text inserted into the computer` was in Courier New, menu navigation was by the pipe symbol, and italic words. The entire section was written in past tense and provided enough detail of what was completed.

New print admin user creation

A new user account was created on G13SRV01VM for the separation of print administration duties across the enterprise forest. The account was placed in the IT_Department OU since print administrator belonged to the Apex Technologies Corporation's IT department.. The newly created user ensured printer-related tasks were handled by an administrative role instead of a full domain administrator.

1. Clicked *Compute* | *VMs* | *G13SRV01VM* | **Launch console** | **Send CtrlAltDelete**
2. Typed `password` in *password* field
3. Selected *Server Manager* | *Tools* | *Active Directory Users and Computers* | *group13.c24200.cit.lcl*
4. Right-clicked *IT_Department*
5. Clicked *New* | *User*
6. Typed `Print` in *first name* field
7. Inputted `Admin` in *last name* field
8. Inserted `padmin` in *user logon name* field
9. Clicked **Next** in *New Object - User* window
10. Typed `password` in *password*
11. Inputted `password` in *confirm password* field
12. Deselected *User must change password at next logon*
13. Selected *User cannot change password* | **Next** *in New Object - User* window | **Finish**

User added to Print Operator group

Elevated privileges were given to the padmin account on G13SRV01VM by the addition of the user to the Print Operators group. The membership allowed control of printer settings, paused active queues, and management of print jobs for all users in the domain. Completion of the configuration on G13SRV01VM domain controller ensured print administrative authority across Apex Technologies Corporation's domain forest.

1. Clicked *IT_Department*
2. Right-clicked *Print Admin*
3. Selected *Properties* | *Member of* | **Add…**
4. Typed `Print Operator` in *Enter object name to select* field
5. Clicked **Check Names** | **OK** in *Select Groups* window | **Apply** | **OK** *in Print Admin Properties* window

General print queue creation

A centralized network printer was implemented on G13SRV01VM for enterprise-level print operations. The setup used the Print and Document Services role and a standard Transmission Control Protocol/Internet Protocol (TCP/IP) port, which linked the logical print server to the printer via the printer's IP address. The configuration supported centralized control of all print traffic across Apex Technologies Corporation's network.

1. Selected *Server Manager* | *Manage* | *Add Roles and Features* | **Next** in *Before you begin* window | **Next** in *Select Installation Type* window | **Next** in *Select destination server* window | *Print and Document Services* | **Add Features**
2. Clicked **Next** in *Select Server Roles* window | **Next** in *Select Features* window | **Install** | **Close**

3. Selected *Tools | Print Management | Print Servers | G13SRV01VM (local)*
4. Right-clicked *Ports*
5. Selected *Add Port… | Standard TCP/IP* | **New Port…** | **Next** in *Add Standard TCP/IP Printer Port Wizard* window
6. Typed `44.3.1.238` in *printer name or IP address* field
7. Clicked **Next** in *Add Port* window | **Finish** | **Close**
8. Right-clicked *Printers*
9. Clicked *Add Printer… | Add a new printer using an existing port*
10. Selected *44.3.1.238 (Standard TCP/IP Port)* | **Next** in *Printer Installation* window | **Next** in *Printer Driver* window | *MS Publisher Color Printer* | **Next** in *Printer Installation* window
11. Typed `Lab242Printer` in *printer name* field
12. Inputted `Lab242Printer` in *share name* field
13. Clicked **Next** in *Printer Name and Sharing* Settings | **Next** in *Printer Found* window | **Finish**

Created second queue for printer that printed on both sides

A secondary print queue was created on G13SRV01VM root domain controller and supported general printing for all users across all departments of Apex Technologies Corporation. The queue used the same hardware and driver as the primary queue and enabled double-sided printing by default for all domain users. Also, the queue's priority was configured to be much lower compared to the non-duplex print queue, which was configured for administrative print operations.

1. Selected *Print Management | Print Servers | G13SRV01VM (local)*

2. Right-clicked *Printers*
3. Clicked *Add Printer… | Add a new printer using an existing port*
4. Selected *44.3.1.238 (Standard TCP/IP Port)* | **Next** in *Printer Installation* window | *Use an existing printer driver on this computer | MS Publisher Color Printer* | **Next** in *Printer Driver* window
5. Typed `Group13Duplex` in *printer name* field
6. Inputted `Group13Duplex` in *share name* field
7. Clicked **Next** in *Printer Name and Sharing Settings* window | **Next** in *Printer Found* window | **Finish**

Limited non-duplex printer to only members of the administrators' group

Permission was restricted on the standard non-duplex printer on G13SRV01VM, which prevented unauthorized usage of simplified print resources. Security permissions removed the Everyone group and left the queue access only for administrators and print operators. Queue priority was raised to 99, which placed administrative print jobs above duplex print jobs in the print queue.

1. Selected *Server Manager | Tools | Print Management | Print Servers | G13SRV01VM (local) | Printers*
2. Right-clicked *Lab242Printer*
3. Clicked *Properties… | Security | Group or user names | Everyone* | **Remove** | **Apply** | *Advance*
4. Typed `99` in *Priority* field
5. Clicked **Apply** | **OK** in *Lab242Printer Properties* window

Configured PowerShell Remoting Trusted Hosts on G13SRV05

PowerShell remoting was enabled for G13SRV05 so connections from other servers in the domain forest were allowed. One of the commands included a list of trusted hosts as authorized domain servers. The configuration allowed centralized control of domain services across the trusted hosts without the need for direct console interaction.

1. Clicked *Prism Central Dashboard* | *Compute* | *VMs* | *G13SRV05* | **Launch console** | *Send CtrlAltDelete*
2. Typed `password` in *password* field
3. Selected *Windows PowerShell*
4. Typed `Enable-PSRemoting -Force` in *terminal*
5. Pressed **Enter**
6. Inputted `Get-Item WSMan:\localhost\Client\TrustedHosts`
7. Inserted `Set-Item WSMan:\localhost\Client\TrustedHosts -Value "G13SRV01VM,G13SRV03VM,G13SRV04,G13SRV05,G13SRV06,G13SRV07" -Force`
8. Typed `Get-Item WSMan:\localhost\Client\TrustedHosts`

Initialized and validated PowerShell Remote Session on G13SRV01VM

A remote session started on G13SRV01VM through PowerShell. Domain administrative credentials opened a PowerShell remote session for access to G13SRV05 without any issues. The test showed Windows Remote Management (WinRM) worked correctly, and the security handshake between trusted hosts worked as expected.

1. Clicked *Prism Central Dashboard* | *Compute* | *VMs* | *G13SRV01VM* | **Launch console** | *Send CtrlAltDelete*
2. Typed `password` in *password* field
3. Selected *Windows PowerShell*
4. Typed `Get-Service WinRM` in *terminal*
5. Pressed **Enter**
6. Inputted `Test-WSMan G13SRV05`
7. Inserted `$cred = Get-Credential`
8. Typed `GROUP13\Administrator` in *username* field
9. Inputted `password` in *password* field
10. Clicked **OK** in *Windows Powershell Credential request* window
11. Typed `Enter-PSSession -ComputerName G13SRV05 -Credential $cred`
12. Inputted `hostname`
13. Inserted `Get-Service | Select-Object -First 5`
14. Typed `Exit-PSSession`

Stopped and Started Print Spooler Service with PowerShell Remoting

G13SRV05's Print Spooler service was stopped and started remotely from G13SRV01VM through the PowerShell remote session. Administrative control was proven to be functional for trusted hosts from the root domain controller. Successful remote access confirmed Apex Technologies Corporation's infrastructure supported centralized control from a single server on the domain.

1. Typed `Enter-PSSession -ComputerName G13SRV05 -Credential $cred`
2. Pressed **Enter**
3. Inserted `Get-Service -Name Spooler`
4. Inputted `Stop-Service -Name Spooler`
5. Typed `Get-Service -Name Spooler`
6. Inserted `Start-Service -Name Spooler`
7. Inputted `Get-Service -Name Spooler`
8. Typed `Exit-PSSession`

Installed DFS Namespace and DFS Replication Roles

The DFS Namespaces and DFS Replication roles were installed on G13SRV01VM via Server Manager for distributed file access across all Active Directory sites. A namespace G13DFS was created under the group13.c24200.cit.lcl domain for a domain-wide share path for roaming profiles and folder redirection data. RoamingProfiles and FolderRedirection folders were created inside the namespace as the storage of user data across Apex Technologies' domain forest.

1. Clicked *Server Manager* | *Manage* | *Add Roles and Features* | **Next** in *Before you begin* window | **Next** in *Select installation type* window | **Next** in *Select destination server* window | *File and Storage Services* | *File and iSCSI Services*
2. Selected *DFS Namespaces* | **Add Features** | *DFS Replication* | **Next** in *Select server roles* window | **Next** in *Select features* window | **Install** | **Close** | *Tools* | *DFS Management* | *Namespaces*
3. Clicked *New Namespace…*
4. Typed `G13SRV01VM` in *server name* field
5. Clicked **Next** in *Namespace server* window
6. Inputted `G13DFS` in *name* field
7. Clicked **Next** in *Namespace name and settings* window | *Domain-based namespace* | **Next** in *Namespace Type* window | **Create** | **Close** | *File Explorer*
8. Typed `\\G13SRV01VM.group13.c24200.cit.lcl\G13DFS` in *address* field
9. Clicked *New* | *Folder*
10. Inputted `RoamingProfiles` in *name* field
11. Clicked *New* | *Folder*

12. Inserted `FolderRedirection` in *name* field

GPO adjustments for DFS namespace share

The Roaming User Profiles and Folder Redirection GPOs were updated G13SRV01VM, so redirected user data were routed to the correct DFS namespace. The route paths were changed \\G13SRV01VM.group13.24200.cit.lcl\G13DFS. The updated policies ensured domain users received profile data and redirected Desktop and Documents through the namespace regardless of the logon location.

1. Selected *Server Manager* | *Tools* | *Group Policy Management* | *Forest: group13.c24200.cit.lcl* | *Domains* | *group13.c24200.cit.lcl*
2. Right-clicked *Roaming User Profiles* | *Edit*
3. Clicked *Computer Configuration* | *Policie*s | *Administrator Templates* | *System* | *User Profiles* | *Set roaming profile path for all users logging onto this computer*
4. Typed `\\G13SRV01VM.group13.c24200.cit.lcl\G13DFS\RoamingProfiles\%USERNAME%` in *roaming profile path* field
5. Clicked **Apply** | **OK** in *Set roaming profile path for all users logging onto this computer* window
6. Right-clicked *Folder Redirection* | *Edit*
7. Selected *User Configuration* | *Policies* | *Windows Settings* | *Folder Redirection* | *Desktop* | *Properties*
8. Typed `\\G13SRV01VM.group13.c24200.cit.lcl\G13DFS\FolderRedirection` in *root path* field

9. Clicked **Apply** in *Desktop Properties* window | **Yes** | **OK** in *Desktop Properties* window | *Documents*
10. Typed `\\G13SRV01VM.group13.c24200.cit.lcl\G13DFS\FolderRedirection` in *root path* field
11. Clicked **Apply** in *Document Properties* window | **Yes** | **OK** in *Document Properties* window

Implemented DFS replication for roaming profiles between sites

DFS replication was configured for the RoamingProfiles share between G13SRV01VM and G13SRV04 with synchronization of roaming profile data across both Active Directory sites. The replication group was named RoamingProfiles-DFS-R used a full mesh topology with full bandwidth selected. The configuration ensured the data of roaming profiles were consistent upon replication between the default and LAN1913 sites.

1. Selected *Server Manager* | *Tools* | *DFS Management* | *Replication*
2. Right-clicked *Replication*
3. Clicked *New Replication Group…* | *Multipurpose replication group* | **Next** in *Replication Group Type* window
4. Typed `RoamingProfiles-DFS-R` in *replication group name* field
5. Clicked **Next** in *Name and Domain* window | **Add**
6. Typed `G13SRV01VM` in *Enter the object name to select* field
7. Clicked **Check Names** | **OK** in *Select Computers* window | **Add**
8. Typed `G13SRV04` in *Enter the object name to select* field

9. Clicked **Check Names** | **OK** in *Select Computers* window | **Next** in *Replication Group Members* window | *Full mesh* | **Next** in *Topology Selection* window | *Replicate continuously using the specified bandwidth* | *Bandwidth* | *Full*
10. Selected **Next** in *Replication Group Schedule and Bandwith* window | **Add…**
11. Typed `C:\Shares\RoamingProfiles` in *local path* field
12. Clicked **OK** in *Add Folder to Replicate* window | **Next** in *Folders to Replicate* window | *G13SRV04* | **Edit…** | *Enabled*
13. Typed `C:\Shares\RoamingProfiles` in *local path of folder* field
14. Clicked **OK** in *Edit* window | **Next** in *Local Path of RoamingProfiles on Other Members* window | **Create** | **Close** | **OK** in *Replication Delay* window

Implemented DFS replication for folder redirection between sites

DFS replication was configured for the FolderRedirection share between G13SRV01VM and G13SRV04 with synchronization of redirected folders data across both Active Directory sites. The replication group was named FolderRedirection-DFS-R used a full mesh topology with full bandwidth selected. The configuration ensured users at LAN1913 site received redirected folder access from G13SRV04 domain controller.

1. Selected *Server Manager* | *Tools* | *DFS Management* | *Replication*
2. Right-clicked *Replication*
3. Clicked *New Replication Group…* | *Multipurpose replication group* | **Next** in *Replication Group Type* window
4. Typed `FolderRedirection-DFS-R` in *replication group name* field
5. Clicked **Next** in *Name and Domain* window | **Add**
6. Typed `G13SRV01VM` in *Enter the object name to select* field

7. Clicked **Check Names** | **OK** in *Select Computers* window | **Add**
8. Typed `G13SRV04` in *Enter the object name to select* field
9. Clicked **Check Names** | **OK** in *Select Computers* window | **Next** in *Replication Group Members* window | *Full mesh* | **Next** in *Topology Selection* window | *Replicate continuously using the specified bandwidth* | *Bandwidth* | *Full*
10. Selected **Next** in *Replication Group Schedule and Bandwith* window | **Add…**
11. Typed `C:\Shares\FolderRedirection` in *local path* field
12. Clicked **OK** in *Add Folder to Replicate* window | **Next** in *Folders to Replicate* window | *G13SRV04* | **Edit…** | *Enabled*
13. Typed `C:\Shares\FolderRedirection` in *local path of folder* field
14. Clicked **OK** in *Edit* window | **Next** in *Local Path of FolderRedirection on Other Members* window | **Create** | **Close** | **OK** in *Replication Delay* window

G13SRV07 Deployment

The G13SRV07 VM was deployed on the G13NUTXSRV01 from the G13WIN2025TEMP template. The server was deployed for update management of Windows Server 2025 servers and Windows 11 clients in the domain with WSUS. A new 500 Gigabytes (GB) virtual hard drive was attached for storage of the updates pulled from clients across the domain. The deployment of G13SRV07 allowed for the configuration of WSUS for servers across the group13.c24200.cit.lcl domain.

1. Clicked *Compute* | *Templates* | *G13WIN2025TEMP* | **Deploy VMs**
2. Typed `G13SRV07` in *Name* field

3. Clicked *Cluster* | *G13NUTXSRV01* | **Next** in *Configuration* window | *Networks* | *G13NUTXSRV01* | **Trash icon** | **Attach to Subnet** | **Save** in *Attach to Subnet* window | **Next** in *Resources* window | **Next** in *Management* window | **Deploy**
4. Selected *Compute* | *VMs* | *G13SRV07* | **Update** | **Next** in *Configuration* window | *Attach Disk* | *Storage Container* | *G13VM* | *Bus Type* | *SATA*
5. Typed `500` in *Capacity* field
6. Clicked **Save** in *Attach Disk* window | **Save** in *Update VM* window | **More** | *Power On* | **Launch Console** | Send *CtrlAltDelete*
7. Typed `password` in *password* field
8. Clicked **Accept** | *Settings* | *System* | *Rename*
9. Typed `G13SRV07` in *name* field
10. Clicked **Next** in *Rename your PC* window | **Restart now**

G13SRV07 further configurations

After deployment, extra network settings were completed on G13SRV07. The server received a static Internet Protocol version 4 (IPv4) address and was joined to the group13.c24200.cit.lcl domain (See *Appendix B: IP Addresses*). The configurations allowed communication with clients in the domain and updates to be pulled to WSUS from Windows Server 2025 and Windows 11 servers across the domain.

1. Typed `password` in *password* field
2. Selected *Control Panel* | *Network and Sharing Center* | *Change adapter settings* | *Ethernet* | *Properties*
3. Deselected *Internet Protocol Version 6 (IPv6)*

4. Clicked *Internet Protocol Version 4 (TCP/IPv4)* | **Properties** | *Use the following IP address*
5. Typed `44.38.13.34` in *IP Address* field
6. Inserted `255.255.255.0` in *Subnet Mask* field
7. Inputted `44.38.13.1` in *Default Gateway* field
8. Typed `127.0.0.1` in *Preferred DNS Server* field
9. Inserted `44.38.13.27` in *Alternate DNS Server* field
10. Clicked **OK** in *Internet Protocol Version 4 (TCP/IPv4) Properties window* | *Settings* | *System* | *About* | *Domain or Workgroup* | **Change…** | *Member of* | *Domain*
11. Typed `group13.c24200.cit.lcl` in *domain name* field
12. Clicked **OK** in *Computer Name/Domain Changes* window
13. Inputted `GROUP13\Administrator` in *username* field
14. Inserted `password` in *password* field
15. Clicked **OK** in *Computer Name/Domain Changes* window | **OK** in *Welcome to the group13.c24200.cit.lcl domain* window | **Close** in *System Properties* window | **Restart Now**

WSUS role installation

The WSUS setup on G13SRV07 required configuration of the secondary storage drive. The dedicated D: drive was partitioned and provided a separate location for the update store, which prevented full capacity to be reached. After the drive was formatted, the WSUS role was installed through Server Manager and allowed further configuration of WSUS via the setup wizard.

1. Typed `password` in *password* field
2. Inputted `Create and format hard disk partitions` in *Windows search* field
3. Clicked *Open* | **OK** in *Initialize disk* window
4. Right-clicked *499.98 GB Unallocated*
5. Selected *New Simple Volume…* | **Next** in *New Simple Volume Wizard* window | **Next** in *Specify Volume Size* window | **Next** in *Assign Drive Letter or Path* window
6. Typed `WSUS` in *volume label* field
7. Clicked **Next** in *Format Partition* window | **Finish** | *File Explorer* | *This PC* | *WSUS (D:)* | **New Folder**
8. Inputted `WSUS` in *folder name* field
9. Selected *Server Manager* | *Manage* | *Add Roles and Features* | **Next** in *Before you begin* window | **Next** in *Select Installation Type* window | **Next** in *Select destination server* window | *Windows Server Update Services* | **Add Features**
10. Clicked **Next** in *Select server roles* window | **Next** in *Select features* window | **Next** in *Windows Server Update Services* window | **Next** in *Select role services* window
11. Inserted `D:\WSUS` in *path* field
12. Clicked **Next** in *Content location selection* window | **Next** in *Web Server Role (IIS)* window | **Next** in *Select role services* window | **Install** | **Close**
13. Selected *Tasks* | *Launch Post-Installation tasks*

WSUS instance implementation

After the WSUS role was installed on G13SRV07, the WSUS server was configured for synchronization from Microsoft Update, The product was configured so only Windows 11 and Windows Server 2025 hosts had updates pulled to G13SRV07. The setup ensured only relevant updates were approved and downloaded for Apex Technologies Corporation's Windows servers and clients.

1. Selected *Server Manager* | *Tools* | *Windows Server Update Services* | **Next** in *Before you Begin* window | **Next** in *Join the Microsoft Update Improvement Program* window | *Synchronize from Microsoft Update* | **Next** in *Choose Upstream Server* window
2. Clicked **Next** in *Specify Proxy Server* window | **Start Connecting** | **Next** in *Connect to Upstream Server* window | **Next** in *Choose Languages* window
3. Deselected *Windows*
4. Selected *Microsoft Server Operating System - 24H2* | *Windows 11* | **Next** in *Choose Products* window | **Next** in *Choose Classifications* window | *Synchronize automatically*
5. Typed `3:00:00 AM` in *First synchronization* field
6. Clicked **Next** in *Set Sync Schedule* window | *Begin initial synchronization* | **Next** in *Finished* window | **Finish**

Configured Client WSUS GPO

After the WSUS server came online, a GPO was applied at the root of the group13.c24200.cit.lcl domain for the enforcement of centralized patches. The policy redirected all domain-joined clients to the internal WSUS server on G13SRV07 for update detection and reports instead of the default Microsoft Windows Update. The configuration placed a scheduled download and installation window and pushed security updates across all domain servers and clients without local user involvement.

1. Selected *Server Manager | Tools | Group Policy Management | Forest: group13.c24200.cit.lcl | Domains | group13.c24200.cit.lcl*
2. Right-clicked *Group Policy Objects*
3. Clicked New
4. Typed `WSUS Client Configuration` in *name* field
5. Clicked **OK** in *New GPO* window
6. Right-clicked *WSUS Client Configuration | Edit*
7. Clicked *Computer Configuration | Policies | Administrative Templates | Windows Components | Windows Update | Manage updates offered from Windows Server Update Service | Specify intranet Microsoft update service location | Enabled*
8. Typed `http://G13SRV07:8530` in *Set the intranet update service for detecting updates* field
9. Inputted `http://G13SRV07:8530` in *Set the intranet statistics server* field
10. Clicked **Apply** | **OK** in *Specify intranet Microsoft update service location* window | *Manage end user experience | Configure Automatic Updates | Enabled | Configure automatic updating: | 4 - Auto download and schedule the install* | **Apply**

11. Selected **OK** in *Configure Automatic Updates* window
12. Right-clicked *group13.c24200.cit.lcl*
13. Clicked *Link an Existing GPO…* | *WSUS Client Configuration* | **OK** in *Select GPO* window | *Terminal*
14. Typed `gpupdate /force`
15. Inserted `wuauclt /detectnow`

Windows Admin Center implementation

After the deployment of G13SRV06, Windows Admin Center was configured as the web-based management platform for all servers and clients in group13.c24200.cit.lcl domain. The gateway made the console reachable from both within the domain network and from external environments over port 6600. All domain controllers and Windows 11 client workstations were registered manually in the web console, which transitioned administrative control across Apex Technologies Corporation's infrastructure to a single web-based platform.

1. Clicked *Prism Central Dashboard* | *Compute* | *VMs* | *G13SRV06* | **Launch Console** | *Send CtrlAltDelete*
2. Typed `password` in *password* field
3. Clicked *Windows Admin Center Setup* | **Next** in *Windows Admin Center Setup* window | *I accept these terms and understand the privacy statement* | **Next** in *License Terms and Privacy Statement* window
4. Selected **Next** in *Select installation mode* window | *Generate a self-signed certificate (expires in 60 days)* | **Next** in *Select TLS certificate* window | **Next** in *Automatic updates* window | **Next** in *Send diagnostic data to Microsoft* window | **Install** | *Microsoft Edge*
5. Typed `https://g13srv06.group13.c24200.cit.lcl:6600` in *search* field

6. Inputted `GROUP13\Administrator` in *username* field
7. Inserted `password` in *password* field
8. Clicked **Sign In** | **Add** | **Add manually** | *Servers* | **Add**
9. Typed `G13SRV01VM` in *Server name* field
10. Selected **Add** *the server name exactly as entered* | **Add**
11. Repeated steps 8-10 for *G13SRV03VM*, *G13SRV04*, *G13SRV05*, *G13SRV07*
12. Clicked **Add** | **Add manually** | *Windows PCs* | **Add** | *Search Active Directory*
13. Typed `G13S3WIN11VM` in *server name* field
14. Clicked **Add**
15. Repeated steps 12-15 for *G13S2WIN11*, *G13S1WIN11VM*, G13WIN11

Windows Admin Center made accessible from external environment

The firewall on G13SRV06 was adjusted, so remote management worked from outside the local CIT gateway. An inbound rule was added in Windows Defender Firewall for TCP traffic on port 6600, which is the port used by Windows Admin Center. The configuration ensured the Windows Admin Center management page could be accessed via https://44.38.13.85:6600 in the browser, but still kept normal security controls for all other network traffic.

1. Typed `Windows Defender Firewall with Advanced Security` in *search* field
2. Selected *Inbound rules* | *New Rule…* | *Port* | **Next** in *Rule Type* window | *TCP*
3. Typed `6600` in *port* field

4. Clicked **Next** in *Protocol and Ports* window | **Next** in *Action* window | **Next** in *Profile* window
5. Typed `WAC External Access` in *name* field
6. Clicked **Finish**

Microsoft Windows Server 2025 Activation

The activation of the Microsoft Windows Server 2025 virtual domain controllers allowed the domain forest support administrative abilities and security controls. Configurations were applied across G13SRV07 for all controllers operated at the licensed and compliant level within the Nutanix environment. All operations were performed through the Command Prompt for linkage between the virtual instances and the campus Key Management Service (KMS) for authorized usage. Once the activation commands were in place, the license for the new virtual setup reached completion, and the enterprise environment was able to be completed.

1. Clicked *Prism Central Dashboard* | *Compute* | *VMs* | *G13SRV07* | **Launch console** | *Send CtrlAltDelete*
2. Typed `password` in *password* field
3. Inputted `Command Prompt` in *search* field
4. Clicked *Run as administrator*
5. Inserted `slmgr /ipk TVRH6-WHNXV-R9WG3-9XRFY-MY832`
6. Pressed **OK**
7. Typed `slmgr.vbs /skms kms01.itap.purdue.edu`
8. Inputted `slmgr.vbs /ato`

RESULTS

Apex Technologies Corporation improved printing, updates, and remote management across the domain. A new user named padmin was created and added to the Print Operators group for printer control. A network printer was installed on G13SRV01VM, and two print queues were created: a normal queue and a duplex queue. The duplex queue was opened to all users, while the normal queue remained limited to administrators. After tests, print jobs could be paused, moved, and managed by the print operators. The admin-only queue received a higher priority so urgent jobs printed first.

PowerShell Remoting provided remote administration between the two domain controllers. A remote session from G13SRV01VM to G13SRV05 allowed service checks and service control. The Print Spooler service was stopped and started through the remote session, which proved server management worked without direct console access.

DFS provided data redundancy across two Active Directory sites. A namespace at \\group13.c24200.cit.lcl\dfs gave users one simple path for file access. DFS Replication kept roaming profiles and redirected folders in synchronization between the default site and the LAN1913 site. GPOs were updated so users pulled profile data from the closest server during logon.

WSUS on G13SRV07 centralized update control. A 500 GB virtual hard drive stored approved updates for Windows 11 and Windows Server 2025. A WSUS GPO redirected all domain computers toward the internal update server instead of Microsoft’s public servers.

Windows Admin Center on G13SRV06 provided a single web portal for server and workstation management. Firewall rules allowed access from multiple network segments. All servers and Windows 11 workstations were added to the portal, which enabled real-time monitoring and configurations through a browser.

The physical infrastructure after the continued expansion of Apex Technologies Corporation's systems was done across 44.38.13.0/24 and 44.38.113.0/24 subnets with the two Nutanix CE AHV hosts managed by Prism Central. G13SRV07 VM was added to G13NUTXSRV01 at 44.38.13.34 as the WSUS server and allowed centralized control of update patches for all clients in the system. The network printer 44.2.1.238 was also managed through Print Management on G13SRV01VM with different print queues, which separated administrative and general print jobs across all nine departments. Windows Admin Center on G13SRV06 was implemented for the management of all servers and clients in the system through a local web browser, which reduced the need for direct console access to individual machines.

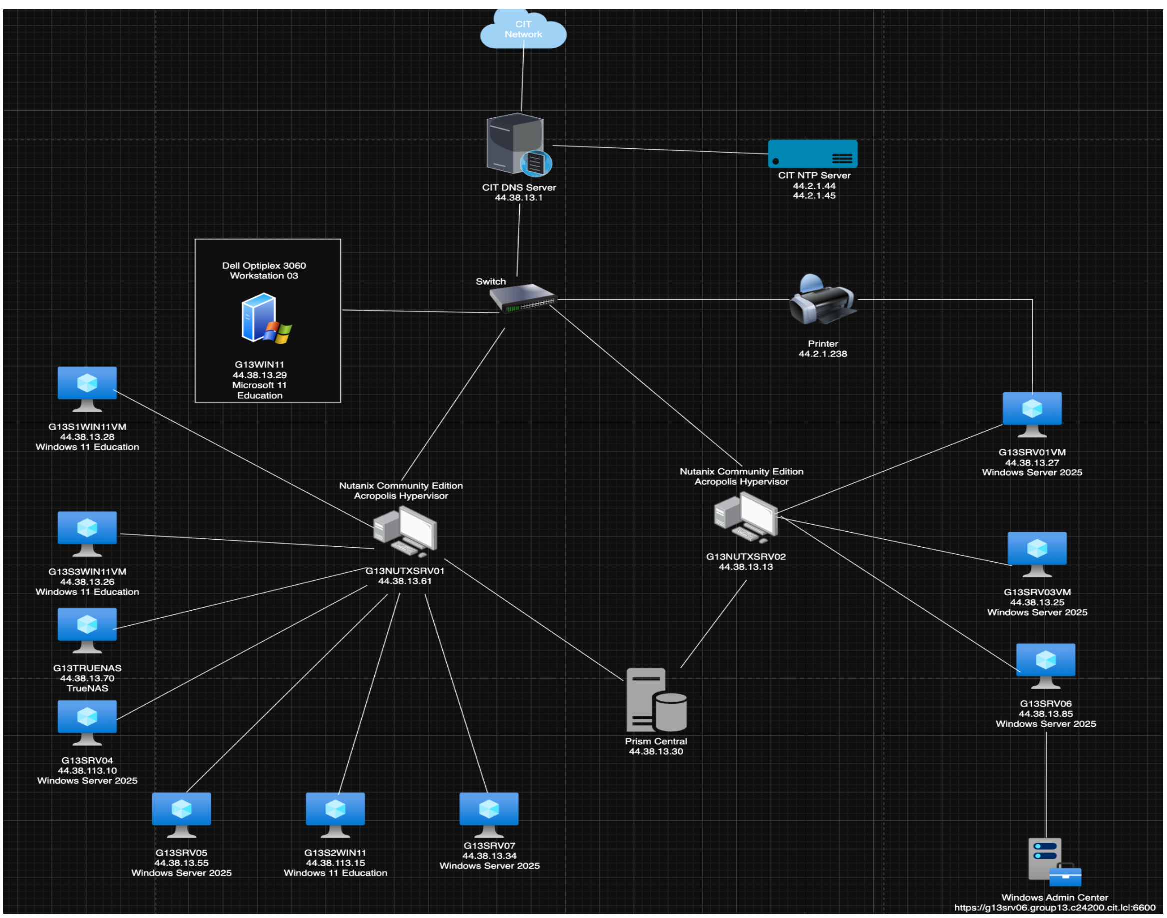

Figure 3: Physical Diagram of Apex Technologies Corporation systems

The updated logical diagram reflected the directory and replication changes made to the group13.c24200.cit.lcl forest after the final expansion campaign of Apex Technologies Corporation's infrastructure. A new user was added to both IT Department OU and Print Operators group, which allowed the authorization of print administration to a member of the IT department. A DFS replication link was established for synchronization of roaming profile and folder redirection data between the two sites. The replication link ensured data consistency for employees of highly authorized IT, Research and Legal departments on the isolated LAN1913 site's servers, and reduced dependency on the primary domain site with lower-level departments.

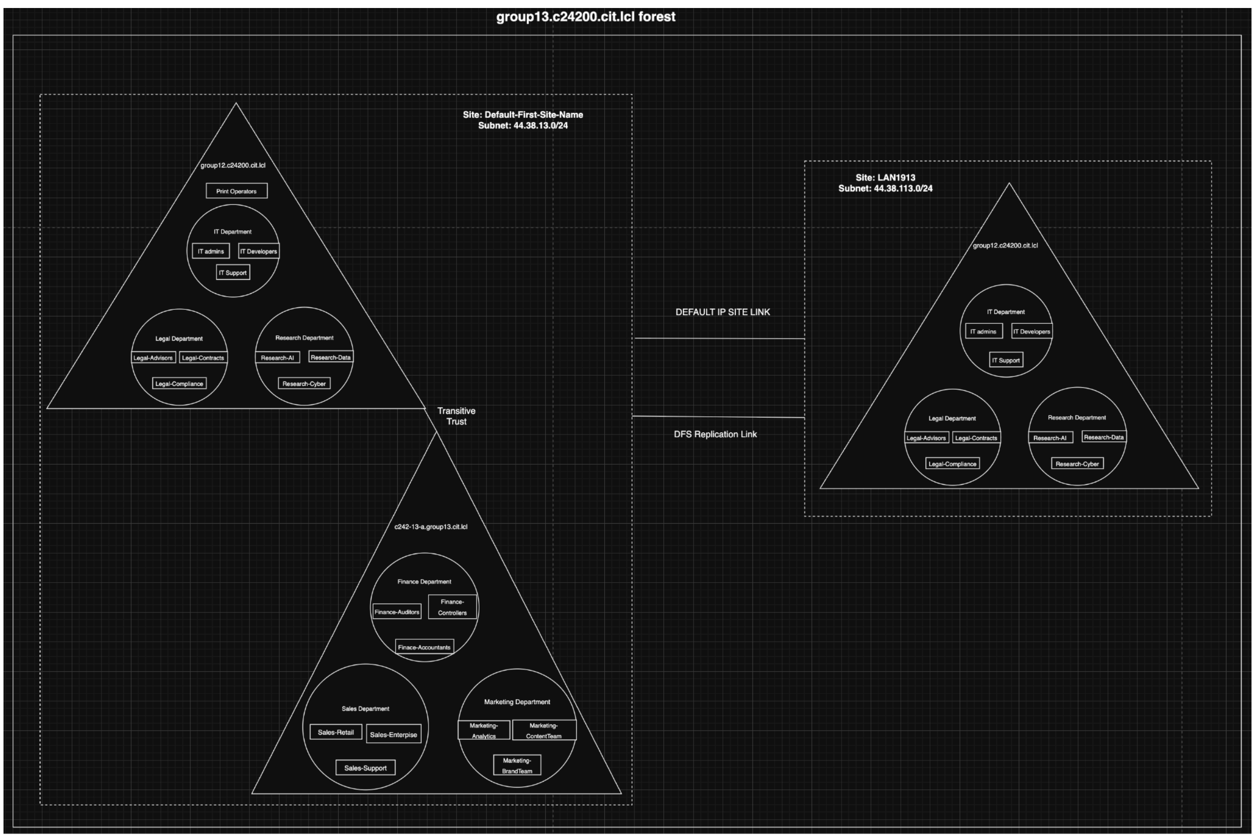

Figure 4: Logical Diagram of Apex Technologies Corporation systems

CONCLUSIONS AND RECOMMENDATIONS

Apex Technologies Corporation's infrastructure expansion met all requirements and expectations. The WSUS server on G13SRV07 addressed the unmanaged update problem and gave the IT department at the LAN1913 site direct control over update patches of all Windows servers across the domain forest. DFS namespaces and replication between two sites resolved the single point of failure problem for roaming profiles and folder redirection. PowerShell remoting was established between servers in the domain and enabled remote administration from G13SRV01VM without direct console interaction on each machine. Windows Admin Center on G13SRV06 allowed browser-based management of all servers and workstations in the domain forest from environments inside and outside of the domain. Print queues managed by Print Operators and administrators separated print administration from general network print across all departments. All requirements reached completion across the Nutanix AHV virtualized environment.

Recommendations

The recommendations reflected lessons from the Apex Technologies Corporation's infrastructure expansion, which included the implementation of WSUS, DFS, PowerShell remoting, and Windows Admin Center across the Apex Technologies Corporation virtualized environment. Each recommendation addressed a technical challenge encountered and provided guidance for future implementations of the same infrastructure. The recommendations were ordered by the order of tasks performed.

Recommendation 1: In the Products and Classifications settings in the WSUS console, select

Microsoft Server Operating System 24H2 for Windows Server 2025 updates. The name

Windows Server 2025 never appeared in the product list, regardless of how many synchronizations were completed. The reason was Microsoft Server Operating System 24H2 being the correct catalog entry for Windows Server 2025 in the WSUS product catalog. Selecting any other entry resulted in no updates for Windows Server 2025 domain controllers in the domain.

Recommendation 2: When configuring PowerShell remoting, be sure to configure trusted hosts on each server using the Set-Item WSman command before trying to connect remotely. By default, any remote connections will be denied even if the service is running on both machines.

Recommendation 3: Run gpupdate /force on all Windows 11 clients after the WSUS GPO was linked to the domain. The reason was update patches for Windows 11 clients were not pulled to WSUS on G13SRV07, while all Windows Server 2025 servers' updates got pulled successfully. Running the command on each Windows 11 client after GPO deployment ensured the WSUS server could pull all update patches.

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APPENDIX A: PROBLEM SOLVING

The section looked back at the main problems encountered in the expansion of Apex Technologies Corporation's environment. A few services failed during setup, which included a Lightweight Directory Access Protocol (LDAP) bind error in Prism Central, a WinRM trust issue in PowerShell Remoting, DFS taking longer than expected for synchronization between sites, and print limits caused by queue permissions. The troubleshooting steps also dealt with updates not pulled to WSUS, firewall rules blocked remote tools, and connectivity issues with the workstations. Each problem was explained in detail in terms of where the failure showed up, what was attempted, and how the fix brought everything back to normal.

WSUS Post-Installation Task Failure on G13SRV07

Problem Description: After the WSUS role was installed on G13SRV07, post-install tasks failed and prevented the creation of the update directory on the D: drive. The failure stopped the first synchronization for Windows 11 and Windows Server 2025. The WSUS console stayed unconfigured and provided no patches for domain clients.

Possible Solutions: One idea involved the reformat of the 500 GB virtual hard drive in case the file system had corruption. Another idea was a manual grant of NT AUTHORITY\Network Service full control over D:\WSUS. The third idea involved the execution of the installation wizard through PowerShell and avoided graphical interface issues.

Solutions Attempted: Disk initialization was verified, and the 500 GB drive was confirmed as a Simple Volume with the correct letter. After a second failure, a manual WSUS folder was created on the D: and the installer was pointed at the directory. Internet Information

Services (IIS) Manager was reviewed for confirmation of the WSUS administration site being bound to port 8530 and in a started state.

Final Solution: A successful fix came from the execution of the post-installation task through Command Prompt with wsusutil.exe postinstall CONTENT_DIR=D:\WSUS. The utility was run with administrative rights, linked the Structured Query Language (SQL) database with the storage directory, and allowed the first synchronization from Microsoft Update. Distribution of updates for the Apex Technologies Corporation environment continued without further issues.

Workstation Internet Connectivity Failure

Problem Description: Dell Pro Tower Plus Server 02 lost Ethernet internet connection after the outage caused by a storm. Prism Central failed in the browser, and Prism Element for the G13NUTXSRV02 cluster failed, which left no access to cluster management. Gateway 44.38.13.1 could not be pinged, and every external check failed.

Possible Solutions: One idea involved the Ethernet cable being replaced or moved to another Ethernet port, since a failed Ethernet cable could not be ruled out. Another idea was a potential cold boot for the server, which could have left the Nutanix hypervisor successfully booted up. The replacement of the Network Interface Card (NIC) was also considered since the power surge could have ruined the card.

Solution Attempted: The Ethernet cable was replaced, and the gateway ping still gave no response. A cold boot was performed on the physical server, and the cluster was restarted, but the CIT gateway was still unreachable. The command ip addr was run on the Nutanix AHV shell, which showed br0 interface was unavailable. Ultimately, the NIC was identified as the failed component caused by the power surge.

Final Solution: The NIC was replaced by the CIT network administrator. The new NIC restored physical network connectivity on Dell Pro Tower Plus Server 02. The gateway at 44.38.13.1 responded to pings immediately after the replacement. Prism Element and Prism Central both came back online.

PowerShell Remoting Connectivity Error

Problem Description: A remote session from G13SRV01VM to G13SRV05 failed in PowerShell Remoting setup because the WinRM handshake never completed. WinRM services ran on both servers, but the source machine refused the connection since no trust entry existed in the client settings.

Possible Solutions: Several ideas were considered in troubleshooting process. One idea involved the firewall being disabled for verification of blocked ports, but the option was not safe for an enterprise-level setup. Another idea involved a connection attempt with the server's IP address for elimination of DNS problems. Another approach focused on adjustment of the TrustedHosts list, so the source machine recognized the remote server.

Solutions Attempted: WinRM was restarted on both servers. Enable-PSRemoting was ran multiple times for confirmation of the listener's active status. Firewall rules were checked for confirmation of WinRM enabled for the domain profile. None of the steps resolved the issue because the trust settings on the client were still incompleted.

Final Solution: The final fix was the addition of the remote server names to the TrustedHists list with the Set-Item WSMan:\localhost\Client\TrustedHosts -Value "G13SRV01VM,G13SRV03VM,G13SRV04,G13SRV05,G13SRV06,G13SRV07" -Force command. Once the names were added, the WinRM client accepted the connection without any problems. Enter-PSSession command worked as expected afterward, and remote service management across the domain forest became functional

APPENDIX B: IP ADDRESSES

The IP configurations for all VMs and physical workstations deployed within Apex Technologies Corporation's virtualized infrastructure was documented in the tables below for reference. The configurations were manually assigned via each machine's network adapter settings for assurance of successful domain joins. Each table documented the IP address, subnet mask, default gateway, preferred DNS, which was assigned a loopback address, and an alternate DNS which pointed to the domain controller of the target domain. Table 1 documented the IP configurations for G13SRV01VM, Table 2 documented the configurations for G13SRV03VM, Table 3 documented the configurations for G13S3WIN11VM, Table 4 documented the configurations for G13S1WIN11VM, Table 5 documented the configurations for the physical workstation G13WIN11, Table 6 documented the configurations for G13TRUENAS, Table 7 documented the configurations for G13SRV04, Table 8 documented the configurations for G13SRV05, Table 9 documented the configurations for G13SRV06, Table 10 documented the configurations for G13S2WIN11, and Table 11 documented the configurations for G13SRV07.

Table 1: G13SRV01VM IP configurations

IP Configurations of G13SRV01VM with DNS pointed to CIT gateway

IP Address	Subnet Mask	Default gateway	Preferred DNS	Alternate DNS
44.38.13.27	255.255.255.0	44.38.13.1	127.0.0.1	44.2.1.44

Table 2: G13SRV03VM IP configurations

IP configurations of G13SRV03VM with DNS pointed to CIT Gateway

IP Address	Subnet Mask	Default gateway	Preferred DNS	Alternate DNS
44.38.13.25	255.255.255.0	44.38.13.1	127.0.0.1	44.2.1.44

Table 3: G13S3WIN11VM IP configurations

IP configurations of G13S3WIN11VM with DNS pointed to CIT Gateway

IP Address	Subnet Mask	Default gateway	Preferred DNS	Alternate DNS
44.38.13.26	255.255.255.0	44.38.13.1	127.0.0.1	44.2.1.44

Table 4: G13S1WIN11VM IP configurations

IP configurations of G13S1WIN11VM with DNS pointed to CIT Gateway

IP Address	Subnet Mask	Default gateway	Preferred DNS	Alternate DNS
44.38.13.28	255.255.255.0	44.38.13.1	127.0.0.1	44.2.1.44

Table 5: G13WIN11 IP configurations

IP configurations of G13WIN11 with DNS pointed to CIT Gateway

IP Address	Subnet Mask	Default gateway	Preferred DNS	Alternate DNS
44.38.13.29	255.255.255.0	44.38.13.1	127.0.0.1	44.2.1.44

Table 6: G13TRUENAS IP configurations

IP configurations of G13TRUENAS with DNS pointed to CIT Gateway

IP Address	Subnet Mask	Default gateway	Preferred DNS	Alternate DNS

44.38.13.70	255.255.255.0	44.38.13.1	127.0.0.1	44.2.1.44

Table 7: G13SRV04 IP configurations

IP configurations of G13SRV04 with DNS pointed to CIT Gateway

IP Address	Subnet Mask	Default gateway	Preferred DNS	Alternate DNS
44.38.113.10	255.255.255.0	44.38.113.1	127.0.0.1	44.2.1.44

Table 8: G13SRV05 IP configurations

IP configurations of G13SRV05 with DNS pointed to CIT Gateway

IP Address	Subnet Mask	Default gateway	Preferred DNS	Alternate DNS
44.38.13.55	255.255.255.0	44.38.13.1	127.0.0.1	44.2.1.44

Table 9: G13SRV06 IP configurations

IP configurations of G13SRV06 with DNS pointed to CIT Gateway

IP Address	Subnet Mask	Default gateway	Preferred DNS	Alternate DNS
44.38.13.85	255.255.255.0	44.38.13.1	127.0.0.1	44.2.1.44

Table 10: G13S2WIN11 IP configurations

IP configurations of G13S2WIN11 with DNS pointed to CIT Gateway

IP Address	Subnet Mask	Default gateway	Preferred DNS	Alternate DNS
44.38.113.15	255.255.255.0	44.38.113.1	127.0.0.1	44.2.1.44

Table 11: G13SRV07 IP configurations

IP configurations of G13S2WIN11 with DNS pointed to CIT Gateway

IP Address	Subnet Mask	Default gateway	Preferred DNS	Alternate DNS
44.38.13.34	255.255.255.0	44.38.13.1	127.0.0.1	44.2.1.44